

Practical 12 - Development and evaluation of risk prediction models

Survival Analysis

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Datasets required: `rotterdam_p10.csv`

R packages required: `survival`, `pec`, `riskRegression`, `rms`

```
library(survival)
library(pec)
library(riskRegression)
library(rms)
```

Introduction

In this practical we will perform analyses similar to those shown in the lecture, using the Rotterdam breast cancer data. The aim is to develop a prediction model for 5-year risk of the composite outcome of disease recurrence or death from any cause, and to evaluate the predictive performance of the model.

The data used is the same as in Practicals 10 and 11. The variables we will use are summarised in the table below.

Variable name	Description
<code>pid</code>	Patient identifier
<code>year_cat</code>	Year of surgery. 0: before 1990, 1: 1990 or later
<code>age</code>	Age at surgery
<code>meno</code>	Menopausal status. 0=premenopausal, 1=postmenopausal
<code>size</code>	Tumour size: categorical with levels ≤ 20 mm, 20 – 50mm, > 50 mm
<code>grade</code>	Differentiation grade: categorical with levels 2 and 3

Variable name	Description
<code>enodes</code>	Number of positive lymph nodes, transformed as <code>enodes=exp(-0.12*nodes)</code>
<code>pgr</code>	Progesterone receptors (fmol/l)
<code>er</code>	Estrogen receptors (fmol/l)
<code>chemo</code>	Chemotherapy: 0=no, 1=yes
<code>hormon</code>	Hormonal treatment: 0=no, 1=yes
<code>time</code>	Years to the first of recurrence or death without recurrence
<code>outcome</code>	Indicator of composite outcome: 0=censored, 1=event
<code>outcome_cr</code>	Competing events outcome: 0=censored, 1=recurrence, 2=death without recurrence

Aims

The aims of this session are:

- Fit prediction models and evaluate their performance.
- Evaluate prediction model performance using measures of calibration, discrimination, and overall performance.
- Use random split-sampling and k-fold cross-validation for evaluation of prediction model performance.

Question 1

- Load the data.
- It turns out to be sensible to change the scale of the variables `pgr` and `er`, so we will divide each of these by 100 for this analysis.
- Create a training set on which the model will be fitted, and a test set on which the predictive performance of the model will be evaluated. The training set is a random sample of 70% of the cohort, and the remaining 30% for the test set.

```
#load the data
rotd<-read.table("datasets/rotterdam_p10.csv",sep=" ",header=T)

#divide pgr and er by 100
rotd$pgr100<-rotd$pgr/100
rotd$er100<-rotd$er/100

#Randomly split the data into training and test sets
#Use the same random seed so that we all get the same answer
```

```
set.seed(1972)

samp<-sample(rott$pid, 2087, replace=F) #0.7*2982=2087.4

train<-rott[rott$pid%in%samp,]
test<-rott[!rott$pid%in%samp,]
```

Question 2

- Using the training set, fit a Cox regression for the composite outcome using the following predictors: `hormon, chemo, age, meno, size, grade, enodes, pgr100, er100, year_cat`. Use the `x=TRUE` option in `coxph`, as this may be helpful for some of the functions we will use later on.
- Use the model to obtain estimates of 5-year risk for individuals in the test set (e.g. using `predictSurvProb`).
- What is the range of the predicted risks in the test set?

Question 3

In this question we will look at two measures of calibration: mean calibration and the calibration slope. These are obtained for the test data. The code is given, but you can probably do all of this without looking at it.

Mean calibration

- Calculate the mean estimated 5-year risk in the test data, and compare this with the Kaplan-Meier estimate of risk up to 5 years.
- What is the ratio of the observed to the expected risk at 5 years?
- The solution shows how to obtain a 95% confidence interval for this ratio.

Calibration slope

- Calculate the linear predictor from the risk model for each individual.
- Perform a Cox regression of the outcome on this linear predictor.
- What is the resulting coefficient? This is the calibration slope.

```
#---
#MEAN CALIBRATION

#mean 5-year risk
exprisk.test<-mean(risk.test)
exprisk.test
```

```

#Kaplan-Meier estimate of 5-year risk
km.test<-survfit(Surv(time,outcome)~1,data=test)
obsrisk.test<-1-summary(km.test,times = 5)$surv
obsrisk.test

#ratio of observed/expected risk
OERatio.test<-obsrisk.test/exprisk.test
OERatio.test

#---
#CALIBRATION SLOPE

#linear predictor
lp.test<-predict(cox.pred,newdata=test,type="lp")

#Cox model including lp.test
calib.test<- coxph(Surv(time,outcome) ~ lp.test, data=test)
summary(calib.test)
confint(calib.test)

```

Question 4

Follow the steps below to create a calibration plot:

- Divide the predicted risks in the test data into 10 equal-sized groups (by placing cut-points at the deciles).
- Calculate the mean of the predicted risks for the people in each of the 10 groups.
- Obtain the Kaplan-Meier estimate of 5-year risk in each group (and also the 95% confidence bounds if you like).
- Plot the Kaplan-Meier estimates against the mean predicted risks.

Discuss what the plot shows.

```

#divide predicted risks into 10 equal sized groups
risk.groups<-cut(risk.test,
                 breaks=quantile(risk.test,probs=seq(0,1,0.1)),
                 include.lowest=T,labels = F)

#mean predicted risk in each group
riskpred.group.test<-sapply(1:10,
                             FUN=function(x){

```

```

        mean(risk.test[risk.groups==x])
      })

#observed risk in each group, plus confidence intervals
obsrisk.group.test<-rep(NA,10)
obsrisk.group.test.lowerci<-rep(NA,10)
obsrisk.group.test.upperci<-rep(NA,10)
for(i in 1:10){
  km.test.temp<-survfit(Surv(time,outcome)~1,data=test[risk.groups==i,])
  obsrisk.group.test[i]<-1-summary(km.test.temp,times = 5)$surv
  obsrisk.group.test.lowerci[i]<-1-summary(km.test.temp,times = 5)$lower
  obsrisk.group.test.upperci[i]<-1-summary(km.test.temp,times = 5)$upper
}

#plot observed versus predicted risk
plot(riskpred.group.test,obsrisk.group.test,type="b",
     xlim=c(0,1),ylim=c(0,1),
     xlab="Predicted",ylab="Observed",lwd=2)
for(i in 1:10){
  segments(riskpred.group.test[i],obsrisk.group.test.lowerci[i],
           riskpred.group.test[i],obsrisk.group.test.upperci[i])
}
abline(0,1,col="red", lwd = 2)

```

Question 5

In this question we create a smooth version of the calibration plot in Question 4. Use the code below to implement the following steps (all in the test data):

- Calculate $Z_i = \log\{-\log(1 - \widehat{R}_i(5))\}$ for each person in the test data, where $\widehat{R}_i(5)$ is the estimated 5-year risk.
- Perform a Cox regression including Z_i as the only covariate, using a spline (this is done using a restricted cubic spline in the code). This is called the ‘refitted model’.
- Obtain estimates of risk up to 5 years for each person based on the refitted model including Z_i .
- Plot the estimated risks from the refitted model against the estimated risks from the main prediction model.

```

#generate log{-log(1-risk)}
test$loglogpred.test <- log(-log(1 - risk.test))

```

```

#Fit Cox regression of this using a spline:
#we do this here using the rms package (there are several other options)
calibcox.test<-cph(Surv(time,outcome) ~ rcs(loglogpred.test, 3),
                  data = test,x=T,surv=T,y=T)

#survival predictions from this model at time 5
calib.test<-survest(calibcox.test,newdata=test,times=5)

#make plot
calibplot.test<-data.frame(pred=risk.test,obs=1-calib.test$surv,
                          lower=1-calib.test$lower,upper=1-calib.test$upper)

calibplot.test<-calibplot.test[order(calibplot.test$pred),]

plot(
  calibplot.test$pred,
  calibplot.test$obs,
  type="l",
  lty = 1,
  xlim = c(0, 1),
  ylim = c(0, 1),
  lwd = 2,
  xlab = "Predicted risk from developed model",
  ylab = "Predicted risk from refitted model"
)
lines(calibplot.test$pred,
      calibplot.test$lower,
      type="l",
      lty = 2,
      lwd = 2)
lines(calibplot.test$pred,
      calibplot.test$upper,
      type="l",
      lty = 2,
      lwd = 2)
abline(0, 1, lwd = 2, lty = 1, col = 2)

```

Question 6

In this question we look at measures of discrimination.

- Use the code below to obtain estimates of Harrell's c-index and Uno's C-index. See the solution for how to obtain 95% CIs.
- Interpret the estimates.

```
#Harrell's C
Cindex.test<- concordance(Surv(time,outcome) ~ risk.test,
                          data=test,reverse = TRUE)
Cindex.test$concordance

#Uno's C
UnoC.test<- concordance(Surv(time,outcome) ~ risk.test,
                        data=test,timewt = "n/G2",reverse = TRUE)
UnoC.test$concordance
```

Question 7

Lastly we look at the Brier score, which is a measure of overall performance.

- Use the code below to obtain an estimate of the Brier score and the scaled Brier score.
- Interpret the estimates. What does the scaled Brier score compare?

```
#Brier score: using Score from riskRegression package
brier.test <-Score(list("cox" = cox.pred),
                   formula = Surv(time, outcome)~1,
                   data = test,
                   conf.int = TRUE,
                   times = 4.99,
                   #The time should be just before the time horizon of interest
                   #(unless there happens to be an event at that time)
                   cens.model = "km",
                   metrics = "brier",
                   summary = "ipa")

brier.test$Brier$score
```

Question 8

So far we fitted the prediction model in the training data and evaluated the model performance in the test data. A better approach is to use k-fold cross validation. As a reminder, this involves the following steps, for 10-fold cross-validation:

- Divide the full data (i.e. before it was split in to training and test sets) randomly into approximately 10 equal-sized groups ('folds').
- Exclude fold 1 and fit the model in folds 2-10. Then obtain risk predictions in fold 1.
- Exclude fold 2 and fit the model in the remaining 9 folds. Then obtain predictions in fold 2.
- ...and so on, excluding each fold one by one. This results in a cross-validated risk prediction for each individual.
- Measures of predictive performance are then obtained using the cross-validated risk predictions on the full data.

k-fold cross-validation can be done automatically using the `Score` function in the `riskRegression` package as follows. This gives several measures of predictive performance, and enables plots. It doesn't give estimates of the c-index, but these can be obtained using `cindex` in the `pec` package, which uses bootstrap cross-validation rather than k-fold cross-validation.

```
#10-fold cross-validation using Score from riskRegression
eval.cv <- Score(
  object = list("Cox" = cox.pred),
  #More than one model can be listed in object if you want to compare
  formula = Surv(time, outcome) ~ 1,
  data = rott,
  split.method = "cv10",
  B = 1,
  metrics=c("AUC","Brier"),
  summary="ipa",
  plots="calibration",
  times = 4.99
)

plotCalibration(eval.cv)

#bootstrap cross-validation for the C-index using pec
cindex.boot <- cindex(
  object = list("Cox" = cox.pred),
  formula = Surv(time, outcome) ~ 1,
  data = rott,
  splitMethod = "bootcv",
  cens.model = "marginal",
  confInt = T #Doesn't seem to work yet
)
```


Question 9

- Suppose you were happy to proceed with your model on the basis of the above evaluations of predictive performance. Obtain and write down a formula that could be used to obtain a prediction for a new individual.